**Code No: 153AB** **R18**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year I Semester Examinations, April/May - 2023**
**ANALOG AND DIGITAL ELECTRONICS**
**(Common to CSE, IT, ECM, ITE, CE(SE), CSE(CS), CSE(N))**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART – A

**(25 Marks)**

1.a) Give the applications of PN junction diode. [2]
b) Discuss about diode switching times. [3]
c) Discuss about gain bandwidth product in amplifier using BJT. [2]
d) What is thermal runway? [3]
e) Define de morgan laws. [2]
f) Define the pinch-off voltage. Why the name field effect is used for the device FET? [3]
g) Differentiate between encoder and decoder. [2]
h) How Decimal Adder different from Binary adder? [3]
i) What is excitation table? Write the excitation tables for the SR flip flop. [2]
j) What is state assignment? Explain with a suitable example. [3]

## PART – B

**(50 Marks)**

2.a) Define and derive the equation for diffusion capacitance.
b) Explain positive and negative diode clipper circuits. [4+6]

**OR**

3.a) Briefly discuss about PN junction diode and light emitting diode.
b) Discuss about half wave rectifier with and without capacitive filter. [5+5]

4.a) Explain the input and output characteristics of a transistor in CE configuration.
b) Draw a Self-bias circuit and explain its operation. Derive the equation for Stability factor. [5+5]

**OR**

5.a) Explain various methods used for coupling of multistage amplifiers with their frequency response.
b) Draw and explain equivalent circuit of transistor at low frequencies. [6+4]

6.a) Draw the circuit diagram of common drain amplifier and derive expression for voltage Gain using FET .
b) Simplify the following function and realize using universal gates
F (A,B,C) = A'BC' + ABC + B'C' + A'B' [5+5]

**OR**

7.a) Explain the construction and principle of operation of Enhancement mode N-channel MOSFET.

b) Explain the operation of TTL with neat diagram. [5+5]

8.a) Minimise the following Boolean function using K-map and design a logic circuit using NAND gates.

$$F=\Sigma m\ (0,3,4,7,8,10,12,14)+d(2,6)$$

b) Construct a 3*8 decoder using logic gates and its truth table. [5+5]

**OR**

9.a) Express the function (xy+z)(y+xz) in canonical SOP and POS forms.

b) Implement the following Boolean function with a multiplexer.

$$F(A,B,C,D)=\sum(1, 3, 4, 11, 12, 13, 14,15)$$ [5+5]

10.a) Draw and explain the logic diagram of 4-bit ring counter with the help of timing diagrams.

b) Realize D-FF and T-FF using JK-FF. [5+5]

**OR**

11.a) Explain about the universal shift registers.

b) Discuss in detail about various types of ROM. [5+5]

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